Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	Yes, 45 antibodies detected, 18 RRIDs provided Suggestion: RRID:AB_2288064 : anti-Tubulin b 3 (TUBB3 RRID:AB_262133 : Rabbit anti-Tubulin b RRID:AB_300798 : anti-GFP : anti-tubulin b 3 : Tuj-1 RRID:AB_2313773 : anti-tubulin b RRID:AB_887811 : RRID:AB_887733 : anti-Munc13-1 RRID:AB_887774 : : anti-Glial Fibrillary Acidic Protein (GFAP RRID:AB_10013382 : anti-Glial Fibrillary Acidic Protein : anti-RFP RRID:AB_2209751 : anti-RFP : anti-DDK RRID:AB_2622345 : anti-DDK RRID:AB_2341095 : anti-NeuN : anti-chicken IgY IgG RRID:AB_2340375 : Donkey anti-chicken IgY IgG : anti-rabbit IgG RRID:AB_2340622 : Donkey IgG H+L Alexa : anti-guinea pig IgG RRID:AB_2340476 : IgG H+L Alexa Fluor RRID:AB_10563566 : IgG H+L Alexa Fluor : Anti-chicken IgY : RRID: AB_2636803 Goat RRID:AB_2636803 : Anti-chicken IgY H&L Alexa : IgG H&L Alexa Fluor RRID:AB_2722623 : IgG H&L Alexa Fluor RRID:AB_2534091 : Alexa Fluor Thermo Fisher : AAV1.CMV.PI.EGFP : anti-GFAP : anti-Chicken : anti-Rabbit : bIII tubulin (1:500, TUBB3, 801201, BioLegend), chicken anti-GFP (1:500, ab13970, Abcam) Rabbit anti-Red Fluorescent Protein (RFP, 1:1000, 600-401-379, Rockland), Rabbit anti-RIM1 (1:200, 140 003, Synaptic Systems), Rabbit anti-RIM2 (1:200 140 103, Synaptic Systems) andor mouse anti-DDK (FLAG) (1:500, TA50011-100, OriGene) in PBS containing 10 blocking solution at room temperature for 1 h. : TUBB3 : anti-Red Fluorescent Protein (RFP : FLAG : anti-bIII tubulin	

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
	: Tuj1 : anti-GFP : anti-NeuN : anti-Munc13-1 : RIM2 : anti-VAMP2 : anti-RIM1 Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	Yes, 10 organisms detected, 4 RRIDs provided RRID:IMSR_EM : Munc13-2-- unc13btm1Rmnd RRID:IMSR_EM : Munc13-3-- unc13ctm1Bros RRID:IMSR_JAX : iBOT Suggestion: RRID:IMSR_JAX : C57BL6J : Munc13-1flfl : iBOT B6.FVB-Tg(CAG-boNTB,-EGFP)U75-56FwpJ : RIM1flflRIM2flfl : AAV1.CMV.HI.eGFP-Cre : Munc13-1flfl Munc13-2KOKO Munc13-3KOKO : TeLC.P2A.dTomato Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Provide statement confirming informed consent obtained from study participants.	Not detected.	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Report on age and sex for all study participants.	**Age:**Adult female wild-type (WT, C57BL6J, Charles River) and transgenic mice of both sexes (8-12 weeks old) were used. **Sex:**Adult female wild-type (WT, C57BL6J, Charles River) and transgenic mice of both sexes (8-12 weeks old) were used.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	not detected.	
Randomization	Animals were randomized into treatment and control groups and analyses were conducted blinded to treatment group.	
Blinding	Animals were randomized into treatment and control groups and analyses were conducted blinded to treatment group.	
inclusion/exclusion criteria	Mice with incomplete lesions based on axon morphology and the absence of a GFAP- area were excluded from analysis.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Genes encompassed in the Gene Ontology (GO):0045202 term synapse (n = 1459) were quantified and triplicate experiments were statistically compared at 06h versus 36 h in culture.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not detected.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Rank-rank hypergeometric overlap analysis was conducted with RRHO2 (Cahill et al., 2018) on the datasets to compare gene expression patterns with conditioning (PNL versus Sham), embryonic development (E12.5 versus E17.5), and time in culture (6 h versus 12 h, 24 h, 36 h).	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Materials availability All uniquestable reagents generated in this study are available from the lead contact without restriction.	
If data are publicly available, provide accession number in repository or DOI or URL.	Cre.mCherry SignaGen SL101117 AAV5.hSyn.hM4D(Gi)-mCherry A gift from Bryan Roth Addgene 50475-AAV5 AAV1.CMV-TeNT-LC-P2A-mCherry Sirion Biotech NA Chemicals, peptides, and recombinant proteins HBSS GIBCO 14025-053 HEPES GIBCO 15630-56 DMEMF-12, HEPES Life Technologies 11330032 Fetal bovine serum Thermo Scientific Cat10500064 PenicillinSteptomycin (PenStrep) antibiotics GIBCO 15140122 Collagenase IV Sigma C1889 Trypsin Worthington LS003703 DNase Worthington LS002007 Neurobasal medium GIBCO 12349-015 B-27 supplement GIBCO 17504-044 L-glutamine GIBCO 25030-024 Bovine serum albumin Sigma A3294 Horse serum Pan Biotech P30-0712 Poly-L-lysine Sigma P2636 Laminin Roche 11243217001 Phosphate buffer solution 0.1 M Sigma P5244 Phosphate buffered saline (PBS) AppliChem A0965,9050 Sucrose Fluka 84100 Paraformaldehyde Merck Millipore 104005 Dimethyl sulfoxide (DMSO) Sigma D5879 Fluoromount Sigma F4680-25 Phosphatase inhibitor Roche 04906837001 Protease inhibitor Roche 11836170001 Bradford reagent AppliChem A6932 Ponceau S AppliChem A2935 FastGreen Sigma 68724 Goat serum Invitrogen 16210064 Donkey serum Jackson ImmunoReasearch JIM-017-000-121 Triton X-100 Sigma X100 Baclofen Sigma B5399 Pregabalin Tocris 3775 Critical commercial assays P3 Primary Cell 4D X Kit S (32 RCT) Lonza V4XP-3032 ECL SuperSignal Dura West Thermo Fisher Scientific 10220294 ECL SuperSignal Pico West Thermo Fisher Scientific 15513766 Experimental models: Organismsstrains Mouse: Munc13-1flfl Banerjee et al., 2020 NA Mouse: Munc13-2-- (unc13btm1Rmnd) Varoqueaux et al., 2002 RRID: IMSR_EM:11739 Mouse: Munc13-3-- (unc13ctm1Bros) Augustin et al., 2001 RRID: IMSR_EM:11617 Mouse: RIM1flfl Kaeser et al., 2008 RRID: JAX:015832 Mouse: RIM2flfl Kaeser et al., 2011 RRID: IMSR_JAX:01 5833 Mouse: iBOT Slezak et al., 2012 RRID: IMSR_JAX:018056 Recombinant DNA ef1a-GFP-RIM1a Muller et al., 2020 NA pmaxGFP Lonza CatV4XP-3024 Unc13a (Myc-DDK-tagged) Origene	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
	MR225156 Munc13-1-H567K Betz et al., 1998 NA Munc13-1-DN Shin et al., 2010 NA Munc13-1-W464R Lipstein et al., 2013 NA pCSCMV:tdTomato Waldner et al., 2009 Addgene 30530 Software and algorithms ImageJ NIH, USA NA AxioVision Zeiss NA ZEN Zeiss NA SoftWoRx 3.5.0 Applied Precision NA MetaMorph Microscopy Automation and Image Analysis Software Molecular Devices NA Imaris 9.1 Bitplane NA Other 35 mm glass bottom dish, No. 1.5 Coverslip, 20 mm, Uncoated MatTek CatP35G-1.5-20-C 6-well plates Thermo Scientific Cat140675 4-well plates Thermo Scientific Cat179820 13 mm coverslip Marienfeld Cat01-11530 Immobilon-PSQ PVDF Membrane Millipore CatISEQ00010 X-ray film Thermo Scientific Cat34076 ImmEdge pen Vector laboratories CatH-4000 Fluorescence microscope Zeiss Axio Observer D1 Live cell microscope Applied Precision DeltaVision RT Confocal microscope Zeiss LSM700 2-photon microscope Zeiss LSM 7MP Super-Resolution confocal microscope Zeiss LSM980 Photometrics CoolSnap HQ camera Roper Scientific NA CCD camera Zeiss NA Heating System Ibidi Cat10918 Heating Insert m-Dish 35 mm high for ibidi Heating System Ibidi Cat10934 Gas Incubation System for CO2 Life Imaging Services NA Cryostat Leica CM3050S CaWomat Cawo 2000IR Pneumatic picopump (picospritzer) WPI CatSYS-PV820 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Frank Bradke (frank.bradkedzne.de). Quantification and statistical analysis Transcriptomic analysis Transcriptomic analysis was performed on bulk RNA-Seq of the DRG using a previously published dataset (GEO: GSE66128) (Tedeschi et al., 2016).	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	